

SUBHRAJIT DAS

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EDUCATION

PhD in Computer Science and Engineering Indian Institute of Technology Gandhinagar, Gandhinagar, India <i>Director's PhD Fellow; Advisors: Prof. Abhishek Bichhawat, Prof. Yuvraj Patel</i>	2025 – Present CPI: 9.10
M.Tech. in Computer Science and Engineering Indian Institute of Technology Gandhinagar, Gandhinagar, India <i>Specialization: Computer Systems</i>	2023 – 2025 CPI: 9.84
M.Sc. in Computer Science University of Kalyani, Kalyani, India <i>First Class with Distinction; First Rank in University</i>	2021 – 2023 CGPA: 9.76
B.Sc. (Honours) in Computer Science Panihati Mahavidyalaya, Sodepur, India (<i>Affiliated to WBSU</i>)	2018 – 2021 CGPA: 9.89

RESEARCH INTERESTS

Computer Architecture, Compilers, Distributed Systems, and Usable Privacy & Security

RESEARCH EXPERIENCE

Projects

- Distributing GEMM over Network of Stations [P2]** Oct 2025 – Present
Team Contributor IIT Gandhinagar & University of Edinburgh
- Investigating how a cluster of commodity CPUs can together handle massively data-parallel, compute-intensive tasks such as general matrix multiplication (GEMM), which are typically reserved for GPUs.
- Developer Experience of Consent Management Platform Integration [P1]** Oct 2025 – Present
Team Contributor IIT Gandhinagar & IIT Kharagpur
- Investigating the usability challenges developers face when integrating consent management platforms (CMPs) into web applications.
- Online Authentication Habits of Indian Users [2]** Oct 2023 – May 2024
Team Contributor IIT Gandhinagar
- Conducted a structured survey with 90 Indian participants, analyzing awareness, usage, and perceptions regarding password habits, password managers, and two-factor authentication (2FA). Highlighted many interesting insights, including a tendency to use default settings, and emphasized the need for tailored strategies to enhance password security.

Theses

- Accelerating Large Integer Arithmetic with Parallel Addition, Subtraction, and Vedic-Based Multiplication Using AVX512 [1]** Jan 2024 – Apr 2025
Supervisors: Prof. Abhishek Bichhawat, Prof. Yuvraj Patel M.Tech Thesis, IIT Gandhinagar
- Designed high-performance data-parallel algorithms for large integer addition and subtraction using AVX512, achieving average execution-time speedups of $2\times$ over the GNU Multiple-Precision Arithmetic Library (GMP).
- Studies on Various Maximal Covering Location Problems using Genetic and Artificial Bee Colony Algorithms** Sep 2022 – Jun 2023
Supervisors: Prof. Priya Ranjan Sinha Mahapatra, Dr. Soumen Atta M.Sc Thesis, University of Kalyani
- Designed algorithms to solve the NP-hard probabilistic maximal covering location allocation problem using the artificial bee colony algorithm, matching the optimal benchmark results of the commercial solver CPLEX in 50% of cases, with accuracy comparable to other meta-heuristic models while beating most of them in computational time.

TEACHING EXPERIENCE

- Teaching Assistant** Jul 2023 – Present
Dept. of CSE IIT Gandhinagar
- Assisted with courses such as Distributed Systems and Cloud Computing, Computer & Network Security, Compilers, and Data Structures and Algorithms – I.
- Guest Lectures on SIMD and Code Profiling** Oct 2025
CS 612: Computer Systems IIT Gandhinagar
- Delivered guest lectures on exploiting x86-64 data parallelism using compiler auto-vectorization and intrinsics, and on profiling code with Linux perf and relevant syscalls.

Guest Lecture on RAID

CS 330: Operating Systems

- Delivered a guest lecture on RAID.

Apr 2025

IIT Gandhinagar

Principal Instructor

Dept. of CSE

- Conducted a student-run course (SRC) titled “Code Profiling and Optimization” in collaboration with a colleague, under the mentorship of *Prof. Abhishek Bichhawat*. The course focused on code profiling, performance benchmarking, and leveraging various tools and libraries for code optimization.

Nov 2024

IIT Gandhinagar

ACCEPTED PUBLICATIONS

- [1] **Subhrajit Das**, Abhishek Bichhawat, and Yuvraj Patel, “DigitsOnTurbo: Leveraging SIMD for Accelerating Large-number Arithmetic,” in *32nd ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2027. arXiv: [2604.21566 \[cs.DC\]](#).
- [2] Pratyush Choudhary*, **Subhrajit Das***, Mukul Paras Potta*, Prasuj Das, and Abhishek Bichhawat, “Online Authentication Habits of Indian Users,” in *2024 Conference on Building a Secure & Empowered Cyberspace (BuildSEC)*, 2024, pp. 66–73. DOI: [10.1109/BuildSEC64048.2024.00018](#), **Best Paper Award**.

PREPRINTS & ONGOING WORKS

- [P1] Gayatri Priyadarsini Kancherla, **Subhrajit Das**, Rajdeep Ghosh, Mainack Mondal, and Abhishek Bichhawat, “‘The Most Frustrating Thing I’ve Ever Worked On’: Data-driven Understanding of Developer Challenges in Consent Management Implementation within Web Ecosystem,” **Under review** at the USENIX Security Symposium, 2027.
- [P2] **Subhrajit Das***, Rachel Somerset*, Leping Li, Abhishek Bichhawat, and Yuvraj Patel, “GaaS: GEMM as a Service over a Network of Stations,” **Under review** at the European Conference on Computer Systems (EuroSys), 2027.

ACCEPTED POSTERS

- [W1] **Subhrajit Das**, Abhishek Bichhawat, and Yuvraj Patel, *Complementary Function Co-scheduling Across Call Boundaries*, Poster, Innovations in Compiler Technology (IICT) Workshop, Indian Institute of Science (IISc), Bengaluru, India, 2026.

* denotes equal contribution.

POSITIONS OF RESPONSIBILITY

Volunteer

ACM AsiaCCS 2026

- Volunteered at the 21st ACM Asia Conference on Computer and Communications Security (ACM AsiaCCS).

Jun 1–5, 2026

Bengaluru, India

Volunteer

FSTTCS 2024

- Volunteered at the 44th Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2024), organized by IARCS in association with ACM India.

Dec 16–18, 2024

IIT Gandhinagar, India

Class Representative

Dept. of CSE, MCS

- Facilitated communication between faculty, administration, and batch-mates in matters ranging from class-related concerns to advocating for a fee reduction for the batch.

Oct 2021 – Jun 2023

University of Kalyani

ACHIEVEMENTS

Selected for the Director’s PhD Fellowship at IIT Gandhinagar

Best Paper Award for “Online Authentication Habits of Indian Users” at BuildSEC’24

All India Rank 530 in GATE 2023 (Computer Science)

Awarded UGC NET JRF in Computer Science

Qualified UGC NET in Computer Science for Lecturership

Jan 2025

Dec 2024

Mar 2023

Dec 2022

Dec 2022

SKILLS & INTERESTS

Programming

C, Python, Assembly, Shell, Go, HTML, CSS, JavaScript

Tools & Libraries

Git, SIMD Intrinsics, Linux Perf, GDB, Valgrind, OpenMP, Docker

Hobbies

Music, Traveling

Other Interests

Politics, Geopolitics, Financial Instruments